

**A reusable Bayesian IRT workflow for small-sample instrument validation: A tutorial with
a case study in teacher belief measurement**

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Abstract

Bayesian Item Response Theory (IRT) offers distinct advantages for fitting complex measurement models under sample-size constraints, yet practical guidance on implementing complete Bayesian IRT workflows for applied researchers remains limited. This tutorial presents a reusable workflow for validating multidimensional instruments with limited samples using the R package *brms*. The workflow guides researchers through five integrated steps: (1) preprocessing decisions for sparse ordinal responses, (2) comparing competing dimensional structures including bi-factor models, (3) using discrimination parameters as diagnostic tools for identifying threshold anomalies and misfitting items, (4) selecting among ordinal link functions using cross-validation and stability diagnostics, and (5) incorporating covariates and differential item functioning analysis. Throughout, we emphasise how the rich diagnostic information provided by Bayesian posterior distributions supports informed qualitative judgment at each decision point. The workflow is illustrated through the validation of a 24-item instrument measuring Chinese secondary mathematics teachers' beliefs about equity ($n = 155$). The iterative process identified misfitting items, resolved threshold anomalies, and ultimately supported a bi-factor graded response model with a general inclusive belief factor and three domain-specific components. All analysis code and data are available in a companion repository.

Keywords: Bayesian item response theory; multidimensional IRT; instrument validation; Bayesian workflow; *brms*

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Introduction

Bayesian IRT: A brief overview

Diagnostic toolkit

Case study: Data and setup

Dataset

The case study uses questionnaire data from secondary mathematics teachers ($n = 155$) in public schools across three urban districts in a city in Zhejiang province, China. The three districts are coded 1, 2, and 3 in descending order of socioeconomic status. Three demographic variables were retained for subsequent differential item functioning analysis: teacher gender (male/female), school district (1/2/3), and the year group taught (7/8/9).

Instrument

The instrument comprises 24 items measuring the degree to which teachers hold inclusive, equity-oriented beliefs about mathematics education, as opposed to more traditional or exclusionary orientations. It was adapted from the Teacher Education and Development Study in Mathematics [TEDS-M; Tatto (2013)] with cultural modifications for the Chinese context (e.g., replacing references to “hands-on experiences” with locally recognisable pedagogical reforms such as project-based learning). Items are organised into three subscales:

- *Beliefs about mathematics* (Bm, 8 items) assess whether teachers view mathematics as a fixed body of rules or as a creative, applied discipline. Items endorsing a formalist view are reverse-coded (e.g., “Mathematics is a collection of rules and procedures” [Bm_1_r, reverse-coded]).
- *Beliefs about learners* (Bl, 11 items) assess whether teachers believe mathematical ability is innate and fixed or accessible to all students (e.g., “To be good at mathematics one needs to be talented” [Bl_7_r, reverse-coded]).

- *Beliefs about pedagogy* (Bp, 5 items) assess teachers' receptiveness to innovative and student-centred teaching approaches (e.g., "I'm continuously seeking better ways to teach mathematics" [Bp_3]).

Teachers responded on a 5-point Likert scale from "strongly disagree" to "strongly agree." The full item list is provided in Appendix B.

The instrument's three-dimensional design, grounded in frameworks distinguishing what mathematics is, who can learn it, and how it should be taught (Felton-Koestler, 2017), naturally motivates competing structural hypotheses for IRT modelling.

Competing structural hypotheses

Given the instrument's multidimensional design, we consider three structural hypotheses representing increasingly complex assumptions about the latent space Figure 1. Each hypothesis is grounded in distinct theoretical perspectives on how teacher beliefs are organised.

In the notation below, η_{pi} denotes the linear predictor for person p on item i (mapped to category probabilities by the ordinal link function), β_0 is the fixed intercept, ξ_i is item i 's difficulty, and $\mathbb{I}_{id}$ is a binary indicator assigning item i to dimension $d \in \{M, L, P\}$.

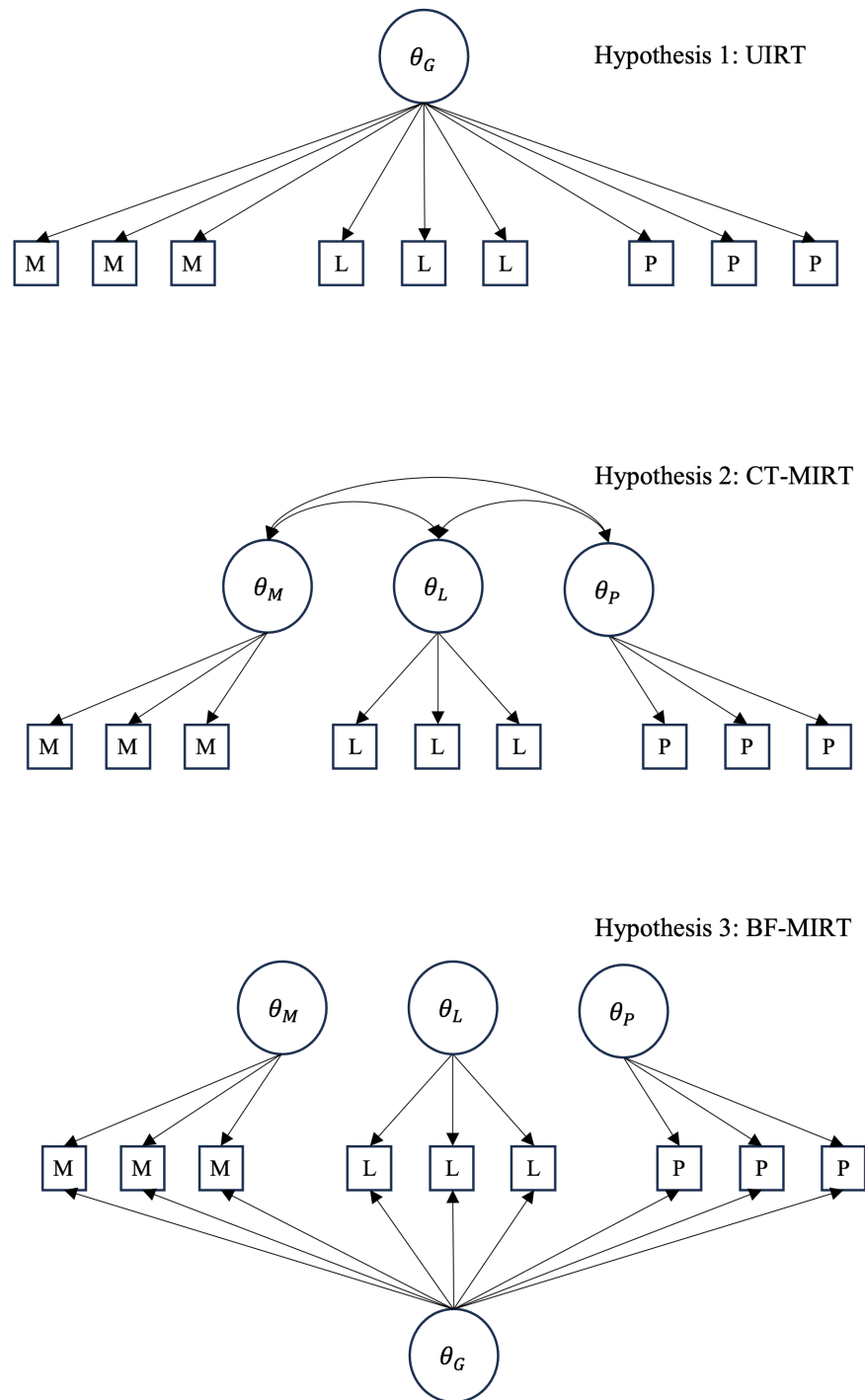
Unidimensional. The simplest assumption is that all items measure a single continuum of inclusive belief. This implies that teachers' orientations toward mathematics, learners, and pedagogy vary in concert, so that a single latent trait θ_p suffices:

$$\eta_{pi} = \beta_0 + \xi_i + \theta_p \quad (1)$$

Correlated traits. According to belief cluster theory (Green, 1971), teachers' beliefs are not monolithic but organised in clusters that may develop and function semi-independently. Applied to this instrument, the three subscales may represent distinct yet interconnected belief clusters, allowing, for instance, a teacher to hold formalist views of mathematics while still believing most students can succeed. Each person is then characterised by a vector $\boldsymbol{\theta}_p = (\theta_{pM}, \theta_{pL}, \theta_{pP})^\top \sim \mathcal{N}(\mathbf{0}, \Sigma)$ with an unrestricted covariance matrix:

Figure 1

Graphical representation of three competing dimensional structures for the equity belief instrument. Top panel: unidimensional model. Middle panel: correlated-traits model. Bottom panel: bi-factor model.



$$\eta_{pi} = \beta_0 + \xi_i + \sum_{d \in \{M, L, P\}} \mathbb{I}_{id} \theta_{pd} \quad (2)$$

Bi-factor. Belief system theories ([Green, 1971](#); [Rokeach, 1968](#)) further propose that beliefs form hierarchical networks in which some core beliefs are stable and broadly influential, while others are more peripheral and context-dependent. Statistically, this implies a bi-factor structure: a general inclusive belief factor captures the shared variance across all items, while dimension-specific residual factors capture variation not explained by the general factor. A general trait θ_p is added alongside orthogonal specific factors θ_{pd}^s :

$$\eta_{pi} = \beta_0 + \xi_i + \theta_p + \sum_{d \in \{M, L, P\}} \mathbb{I}_{id} \theta_{pd}^s \quad (3)$$

The orthogonality constraint $\text{Cov}(\theta_p, \theta_{pd}^s) = 0$ is imposed here.¹ Competing combinations are systematically compared in the workflow later (also see [Appendix A](#) for modelling details).

Computational setup

All models were fitted in R using the brms package ([Bürkner, 2017](#)), which interfaces with Stan ([Carpenter et al., 2017](#)) for Hamiltonian Monte Carlo sampling. Each model used 4 chains with 4,000 iterations per chain (2,000 warmup), yielding 8,000 post-warmup draws for inference. Weakly informative priors were specified following Bürkner (2021): $\mathcal{N}(0, 3)$ for person and item random effect standard deviations, $\mathcal{N}(0, 1)$ for log-discrimination parameters, and LKJ(2) for correlation matrices in the correlated-traits models.² Convergence was assessed by confirming $\hat{R} < 1.01$ for all parameters, adequate bulk and tail effective sample sizes, and no persistent divergent transitions. Additional sampler settings introduced for complex multidimensional models are discussed alongside the relevant workflow steps. Complete code is

¹ It is possible to relax this constraint and estimate correlations between the general and specific factors. However, doing so introduces identification difficulties, as the model must distinguish general-factor variance from correlated specific-factor variance using the same set of item responses. With small samples this also raises the risk of overfitting, where the additional covariance parameters absorb noise rather than capturing meaningful structure.

² The LKJ prior ([Lewandowski et al., 2009](#)) is a distribution over correlation matrices. We use $\eta = 2$ as a mild regularising choice.

provided in the companion repository.

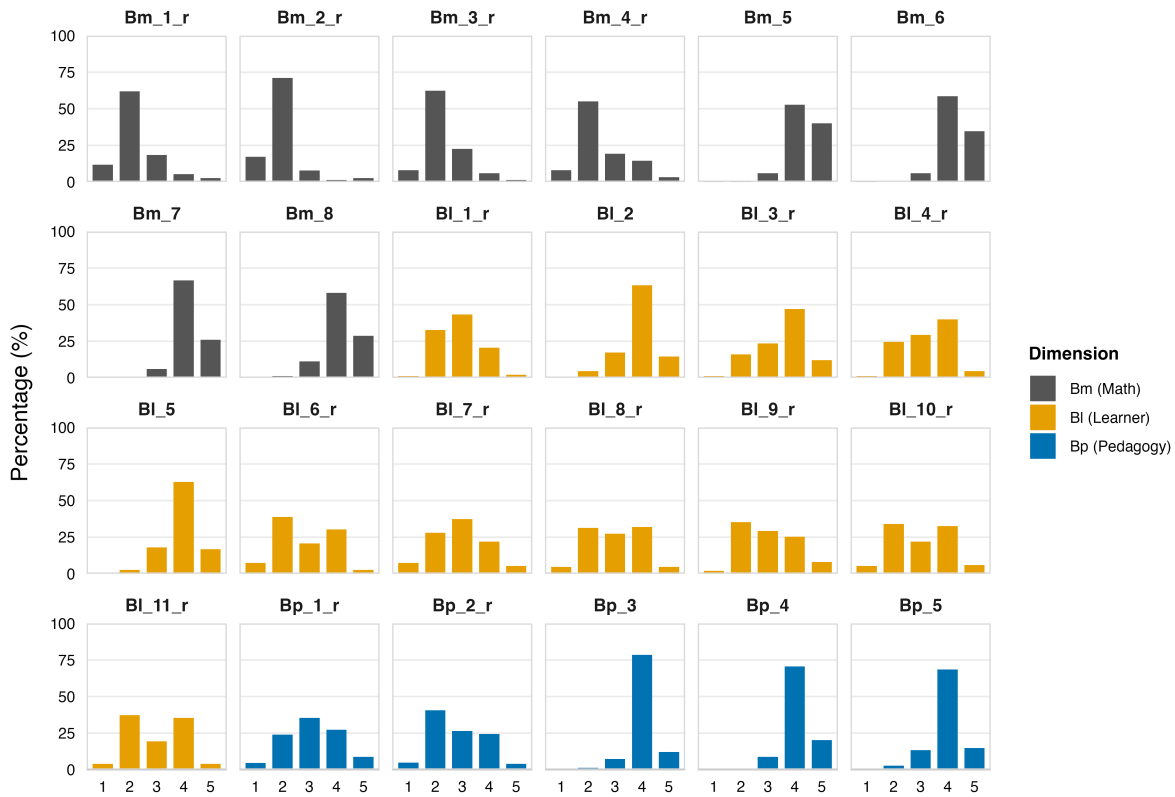
Applying the workflow

Step 1: Preprocessing sparse response categories

We begin by inspecting the observed response distributions. Figure 2 shows the item-level frequency distributions across the original 5-point scale.

Figure 2

Item-level response distributions across the original 5-point scale, coloured by subscale dimension.



A small number of respondents answered only the first few items and left all remaining items blank, likely reflecting disengagement. Because IRT models use full information likelihood, each response contributes independently to estimation regardless of whether the respondent completed the full instrument, so these cases were retained without special treatment.

More consequential is the sparsity of the lowest response category. Category 1 (“strongly disagree”) was very rarely endorsed: twelve of 24 items received fewer than three responses in

this category. When very few respondents select a given category, the model lacks sufficient information to identify the threshold involving that category. Bayesian priors can regularise the estimate to prevent divergence, but the resulting posterior will be prior-dominated rather than data-informed. This is an information problem that no estimation framework can resolve, and it motivates collapsing categories 1 and 2 into a single “disagree” category, reducing the scale from five to four ordered levels. The decision is also justified on semantic grounds: both categories represent forms of disagreement, whereas merging the two middle categories would conflate “disagree” and “neutral,” which are substantively distinct.

Before collapsing, we can first fitted a baseline Rating Scale Model (RSM) to the original 5-point data to inspect the threshold structure. In *brms*, the RSM is specified using `brmsfamily("acat", "logit")`, where *acat* denotes the adjacent-category parameterisation (modelling the odds of each response category relative to its neighbour) and *logit* specifies the logistic link function.

```

null_rsm <- brm(
  formula = response ~ 1 + (1 | item) + (1 | id),
  data = data_1,
  family = brmsfamily("acat", "logit"),
  prior = prior(normal(0, 3), class = "sd")
)

```

Table 1

Threshold estimates from null_rsm

Threshold Transition	Estimate	95% CrI
1: “strongly disagree” to “disagree”	−3.08	[−3.57, −2.61]
2: “disagree” to “neutral”	−0.38	[−0.86, 0.06]
3: “neutral” to “agree”	−0.56	[−1.02, −0.12]
4: “agree” to “strongly agree”	2.16	[1.70, 2.62]

As shown in Table 1, Threshold 1 sits far from the others (−3.08), consistent with the near-absence of “strongly disagree” responses and confirming the need for category collapsing. But the baseline model also reveals a second, less expected pattern: Thresholds 2 and 3 are in

reversed order (-0.38 vs. -0.56), with almost entirely overlapping CrIs. The level of the latent variable required for respondents to move from “disagree” to “neutral” is estimated as higher than that required to move from “neutral” to “agree,” meaning the model cannot meaningfully distinguish these two transitions.

After collapsing categories 1 and 2, we fitted `merge_rsm` with identical model specifications on the 4-category data. We can use `ranef()` to extract item and person parameter estimates and construct a Wright map (Figure 3).

```
ranef <- ranef(merge_rsm)

ranef$item[, , "Intercept"] %>%
  as_tibble(rownames = "item") %>%
  ggplot(aes(x = Estimate, y = item)) +
  geom_point() +
  geom_errorbar(aes(xmin = Q2.5, xmax = Q97.5))

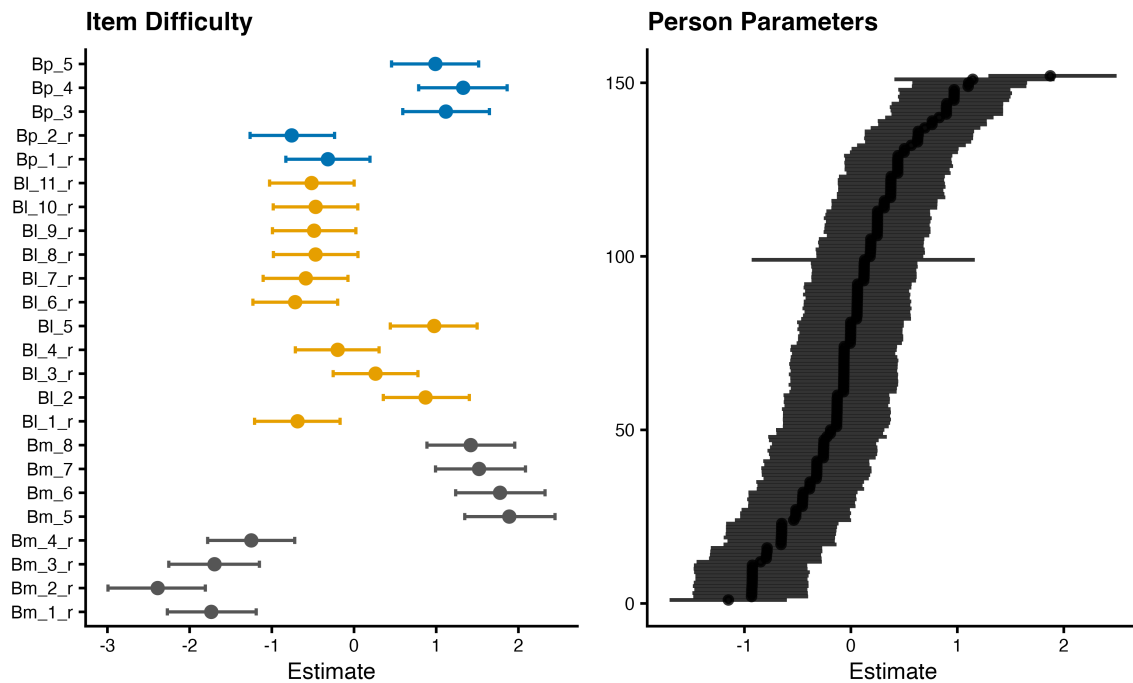
ranef$id[, , "Intercept"] %>%
  as_tibble(rownames = "person") %>%
  arrange(Estimate) %>%
  mutate(id = row_number()) %>%
  ggplot(aes(x = id, y = Estimate)) +
  geom_pointrange(aes(ymin = Q2.5, ymax = Q97.5))
```

The item difficulty map shows a satisfactory distribution of difficulties across the latent continuum, with reasonably sized credible intervals and no abnormally wide estimates. The person ability distribution is approximately normal, though a handful of respondents exhibit very wide credible intervals, likely reflecting the inconsistent response patterns noted earlier. Overall, item-person targeting appears adequate.

The threshold reversal however persists. Thresholds 1 (-0.40 , CrI $[-0.91, 0.1]$) and 2 (-0.56 , CrI $[-1.06, -0.06]$) of `merge_rsm` remain in reversed order with heavily overlapping credible intervals, confirming that the anomaly was not attributable to category sparsity alone. It is worth recalling that the RSM imposes several strong assumptions simultaneously: it assumes a single underlying construct, constrains all items to share the same set of thresholds, and fixes

Figure 3

Wright map from merge_rsm. Left: item difficulty estimates with 95% CrIs. Right: person ability estimates sorted by magnitude.



discrimination to be equal across items. When any of these assumptions is violated in data structure, the shared threshold structure is forced into a compromise that can compress or reverse threshold spacing. The persisting reversal may therefore reflect a more complex dimensionality than the unidimensional model assumes, or it may indicate that certain items discriminate poorly under a constrained equal-discrimination specification, or both. We address these possibilities in turn: Step 2 examines whether a multidimensional structure better accounts for the data, and Step 3 investigates whether freeing item-level discrimination parameters resolves the remaining threshold disorder.

Step 2: Comparing dimensional structures

The belief instrument was designed with three subscales: beliefs about mathematics, beliefs about learners, and beliefs about pedagogy. The unidimensional RSM fitted in Step 1 treats all 24 items as indicators of a single construct, but the persisting threshold reversal suggests

that this may be an oversimplification. If items from different subscales tap into related but distinct aspects of teachers' equity beliefs, forcing them onto a single continuum could distort the shared threshold structure. We therefore compare three competing dimensional specifications, all fitted to the same collapsed 4-category data under the RSM.

The first is the unidimensional model already in hand (`merge_rsm`). The second is a correlated-traits model, which replaces the single person ability term with three dimension-specific traits that are allowed to correlate freely. In `brms`, this is specified by replacing $(1 \mid \text{id})$ with $(0 + \text{dim}_m + \text{dim}_l + \text{dim}_p \mid \text{id})$, where `dim_m`, `dim_l`, and `dim_p` are binary indicators assigning each item to its subscale. The single-bar notation `|` estimates an unrestricted covariance matrix across the three traits. Because correlated random effects introduce additional geometry that the sampler must navigate, we increase `adapt_delta` and `max_treedepth` beyond their defaults, and add an LKJ(2) prior on the correlation matrix to mildly regularise toward lower correlations.

```
ct_rsm <- brm(
  formula = response ~ 1 + (1 | item) +
    (0 + dim_m + dim_l + dim_p | id),
  data = data_2,                      # collapsed 4-category data
  family = brmsfamily("acat", "logit"),
  prior = prior(normal(0, 3), class = "sd") +
    prior(lkj(2), class = "cor"),
  control = list(adapt_delta = 0.99, max_treedepth = 15)
)
```

The third is a bi-factor model, which adds a general factor shared across all items alongside dimension-specific residual factors. A critical implementation detail here is the orthogonality constraint between the general and specific factors. In `brms`, the double-bar notation `||` suppresses the estimation of correlations between random effects. By specifying each factor in a separate `||` term, the general factor and the three specific factors are treated as independent.

```
bf_rsm <- brm(
  formula = response ~ 1 + (1 | item) +
    (1 || id) +
    (0 + dim_m || id) +
    (0 + dim_l || id) +
    (0 + dim_p || id),
  data = data_2,
  family = brmsfamily("acat", "logit"),
  prior = prior(normal(0, 3), class = "sd"),
  control = list(adapt_delta = 0.99, max_treedepth = 15)
)
```

For three models fitted to the same response variable and the same dataset, LOO-CV provides a valid basis for comparison. In `brms`, this is achieved via `loo_compare()`. Table 2 shows that both multidimensional models substantially outperform the unidimensional RSM ($\Delta\text{ELPD} = -69.0$, $\text{SE} = 13.8$), confirming that treating the three subscales as a single latent trait loses meaningful information about the data structure. The correlated-traits and bi-factor models, however, are not distinguishable from each other at this stage ($\Delta\text{ELPD} = -0.2$, $\text{SE} = 2.0$), and both are retained as candidate specifications going forward.

Table 2

LOO-CV comparison of dimensional structures. Models are ordered by ELPD; differences are relative to the best-fitting model.

Model	ELPD difference	SE
bf_rsm (bi-factor)	0.0	
ct_rsm (correlated-traits)	-0.2	2.0
merge_rsm (unidimensional)	-69.0	13.8

Table 3

Threshold estimates from the multidimensional RSMs.

Threshold Transition	bf_rsm (Est. [95% CrI])	ct_rsm (Est. [95% CrI])
1: “disagree” to “neutral”	-0.55 [-1.10, 0.01]	-0.54 [-1.11, 0.05]
2: “neutral” to “agree”	-0.55 [-1.10, 0.01]	-0.54 [-1.11, 0.04]
3: “agree” to “strongly agree”	2.44 [1.88, 3.00]	2.46 [1.89, 3.04]

The multidimensional models do partially improve the threshold structure. Table 3 shows the threshold estimates from both models. The reversal observed in Step 1 is gone, but the first two thresholds remain almost identical with fully overlapping credible intervals under both models, and the model still cannot distinguish the “disagree” to “neutral” and “neutral” to “agree” transitions. Further collapsing these two categories is not a viable solution: “disagree” and “neutral” are semantically distinct responses, and both categories have sufficient observations across items, so merging them would sacrifice meaningful variation in the data without a substantive justification. Accounting for dimensionality has therefore improved but not resolved the threshold anomaly.

Step 3: Freeing discrimination parameters

If we assume that the data quality is reliable, that teachers responded thoughtfully and were able to distinguish between adjacent response categories in a meaningful way, then a reasonable concern is that the model itself imposes assumptions that are too restrictive for the data at hand. For the RSM, a particularly strong assumption is that all items share the same discrimination, fixed at $\alpha = 1$. This becomes problematic when certain items in the instrument exhibit extreme response distributions. Consider Bm_2_r (“Mathematics involves remembering and applying definitions, formulas, mathematical facts and procedures”): nearly all respondents endorsed this item regardless of where they fall on the latent continuum, meaning it contributes very little information about individual differences. When poor items are forced to $\alpha = 1$, the model must adjust the shared threshold parameters to accommodate their universally high endorsement rates. This adjustment compresses the spacing between adjacent thresholds across all items, weakening the model’s ability to distinguish substantive transitions in the response scale (Embretson & Reise, 2013).

To relax this constraint, we only need to add one line to the model specification: $\text{disc} \sim 1 + (1 \mid \text{item})$, which estimates a log-discrimination intercept for each item. The resulting model is a Generalised Rating Scale Model (GRSM).

We fitted both correlated-traits and bi-factor versions (ct_grsm and bf_grsm), and did

LOO-CV comparison across all four models which confirms that the improvement is substantial. For example, `ct_rsm` falls -141.0 ELPD ($SE = 22.4$) below `ct_grsm`, a difference of more than six standard errors.

More importantly for our ongoing diagnostic concern, the threshold anomaly is now resolved. Under `ct_grsm`, Thresholds 1 and 2 are estimated at -0.35 and 0.16 , in the correct monotonic order and with separated credible intervals. The model can now meaningfully distinguish the “disagree” to “neutral” and “neutral” to “agree” transitions.

Figure 4 shows the difficulty and discrimination estimates for both `ct_grsm` (top row) and `bf_grsm` (bottom row).

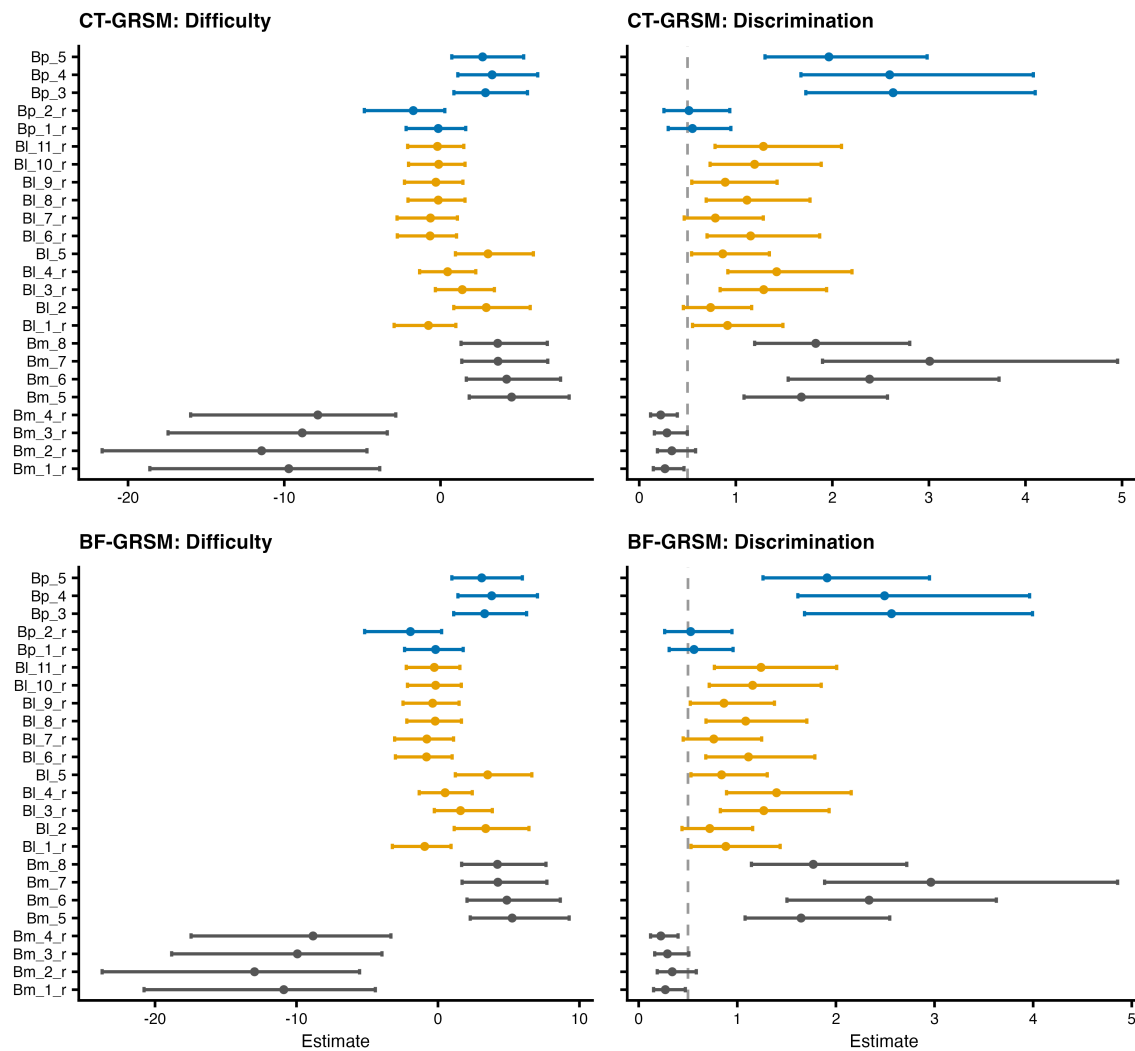
The discrimination panels unsurprisingly reveal substantially different discrimination values across items, confirming that the equal-discrimination assumption was inappropriate. Most items cluster in a reasonable range, but four items stand out: `Bm_1_r`, `Bm_2_r`, `Bm_3_r`, and `Bm_4_r`. These four items all belong to the “beliefs about mathematics” subscale and share the same reverse-coded formalist orientation (e.g., “Mathematics is a collection of rules and procedures”). Their discrimination estimates fall well below the conventional $\alpha = 0.5$ threshold for ‘poor’ items (Embretson & Reise, 2013). This indicates that item responses are virtually independent of the measured latent trait: a respondent’s location on the latent continuum exerts little to no influence on their probability of selecting a given category. Their difficulty estimates are simultaneously problematic with very wide CrIs, meaning they are poorly identified by the data and nearly all respondents find it easy to endorse the exclusive end. While this problematic pattern was invisible under the RSM. Because the equal-discrimination assumption the poor measurement properties were diluted across the shared parameter structure rather than localised to specific items.

Step 4: Item removal and link function comparison

The four problematic items (`Bm_1_r` through `Bm_4_r`) all concern formalist statements about the nature of mathematics (see Appendix B for item wording). These items performed well in the large-scale international survey from which the instrument was adapted (Tatto, 2013),

Figure 4

Item difficulty (left) and discrimination (right) estimates from the *ct_grsm* (top) and *bf_grsm* (bottom). Discrimination values are shown on the exponentiated scale (e^{α}). The dashed line at 0.5 marks a conventional threshold below which items are considered poor discriminators.



suggesting that the poor discrimination observed here reflects a cultural mismatch rather than a universal item defect. Chinese mathematics teachers may not perceive an emphasis on memorisation and procedural fluency as contradicting an emphasis on creativity and application, making these formalist items uninformative for differentiating teachers along the equity belief continuum in this population. Regardless of the explanation, the diagnostic evidence from Step 3 provides sufficient grounds for removal: these items contribute limited and potentially misleading information to the measurement model. After removal, the reduced dataset (`data_3`) contains 20 items.

Comparing ordinal link functions

With dimensionality confirmed and misfitting items removed, we now compare IRT models that differ in their ordinal link function specifications. In addition to the GRSM already fitted, we specify two further model families on the reduced 20-item dataset.

The GPCM (Generalised Partial Credit Model) retains the adjacent-category logit but frees the thresholds to vary across items. This can be achieved by using `(cs(1) | item)`, where `cs()` specifies category-specific effects:

```
ct_gpcm_20 <- brm(
  formula = bf(
    response ~ 1 + (cs(1) | item) +
      (0 + dim_m + dim_l + dim_p | id),
    disc ~ 1 + (1 | item)
  ),
  data = data_3,
  family = brmsfamily("acat", "logit")
)
```

The GRM (Graded Response Model) constrains thresholds to be ordered within each item but uses a different link function entirely. Rather than modelling the odds of each category relative to its neighbour, the GRM models the cumulative probability $P(Y \geq c)$ using a cumulative logit. This requires changing the family specification from "acat" to "cumulative":

```
bf_grm_20 <- brm(
  formula = bf(
    response ~ 1 + (1 | item) +
      (1 || id) + (0 + dim_m || id) +
      (0 + dim_l || id) + (0 + dim_p || id),
    disc ~ 1 + (1 | item)
  ),
  data = data_3,
  family = brmsfamily("cumulative", "logit")
)
```

Model comparison and selection

Crossing these three link function families with two dimensional structures (CT and BF) yields six candidate models, all fitted to the same 20-item, 4-category dataset. Table 4 shows the comparison results.

Table 4

LOO-CV comparison of six candidate models. Models are ordered by ELPD. Δ ELPD and its SE are computed relative to the best-fitting model. $\hat{k} > 0.7$ denotes the number of observations with problematic Pareto $\hat{k}$ diagnostic values. Stacking weights were computed via `loo_model_weights()`.

Model	ELPD (SE)	Δ ELPD (SE)	p_{loo} (SE)	$\hat{k} > 0.7$ (%)	Weight
ct_gpcm_20	−2821.3 (41.8)	0.0	364.4 (12.1)	9 (0.30%)	76.01%
bf_gpcm_20	−2836.5 (41.9)	−15.2 (5.3)	361.4 (11.8)	7 (0.23%)	0.02%
bf_grm_20	−2849.4 (40.7)	−28.1 (10.1)	339.3 (10.3)	1 (0.03%)	23.97%
ct_grm_20	−2851.2 (40.7)	−29.9 (9.9)	342.1 (10.4)	2 (0.07%)	< 0.01%
ct_grsm_20	−2855.3 (41.0)	−34.0 (8.8)	326.1 (10.7)	5 (0.17%)	< 0.01%
bf_grsm_20	−2857.6 (41.0)	−36.3 (8.9)	327.0 (10.7)	3 (0.10%)	< 0.01%

Among the six models, `ct_gpcm_20` achieves the highest ELPD and the largest stacking weight (76%). This is expected: the GPCM allows both discrimination and thresholds to vary freely across items, providing the greatest flexibility. However, this flexibility comes at a cost. The PSIS-LOO diagnostics reveal that the two GPCM models produce the most problematic Pareto $\hat{k}$ values (9 and 7 observations with $\hat{k} > 0.7$, respectively). While these represent a very small proportion of the total (< 0.3%) and are unlikely to substantially alter the overall ELPD

estimates, they indicate that the LOO approximation is unreliable for these specific observations. The two GRM models are notably more stable, with only 1 and 2 problematic observations each.

Both theoretical and practical considerations favour selecting a more stable model. Problematic Pareto $\hat{k}$ values signal that the model is highly sensitive to certain individual observations, meaning the leave-one-out posterior diverges substantially from the full posterior for these cases. As Vehtari et al. (2024) note, this is more likely to occur with less robust model specifications and highly influential observations.

Resolving problematic Pareto $\hat{k}$ values

When problematic Pareto $\hat{k}$ values are detected, the first remedy is moment matching via `moment_match = TRUE`, which adjusts the MCMC draws to better approximate the leave-one-out posterior:

```
loo_bf_grm_20 <- loo(bf_grm_20, moment_match = TRUE)
```

For `bf_grm_20`, moment matching successfully resolved the single problematic $\hat{k}$ value. If moment matching fails to resolve all problematic observations, `reloo = TRUE` can be added, which refits the entire model once for each remaining problematic observation.³

```
loo_ct_grm_20 <- loo(ct_grm_20, moment_match = TRUE, reloo = TRUE)
```

After calibration, both GRM models returned all Pareto $\hat{k}$ estimates below 0.7, and both produced valid MCSE estimates for their ELPD (0.3 for both models). Compared to the ELPD standard error of approximately 40.7, this confirms that the MCMC sampling was more than sufficient. In contrast, when problematic Pareto $\hat{k}$ values ($\hat{k} > 0.7$) remain uncalibrated, the theoretical variance of the importance weights diverges and the MCSE becomes unestimable.

³ Each refit runs the full MCMC sampling procedure. In our case, each refit took approximately 30 minutes, making 2 refits manageable.

Final model selection

bf_grm_20 trailed the top-ranked ct_gpcm_20 by only approximately three standard errors in ELPD and received the second-largest stacking weight (24%), while exhibiting far superior diagnostics. We therefore proceed with bf_grm_20 as the final measurement model. We can also compare the two GRM family models with calibrated LOO estimates to check the final choice.

```
loo_compare(loo_bf_grm_20, loo_ct_grm_20)
#>           elpd_diff se_diff
#> bf_grm_20  0.0         0.0
#> ct_grm_20 -1.7         1.2

loo_model_weights(list(bf_grm = loo_bf_grm_20, ct_grm = loo_ct_grm_20))
#> Method: stacking
#> -----
#>           weight
#> bf_grm 1.000
#> ct_grm 0.000
```

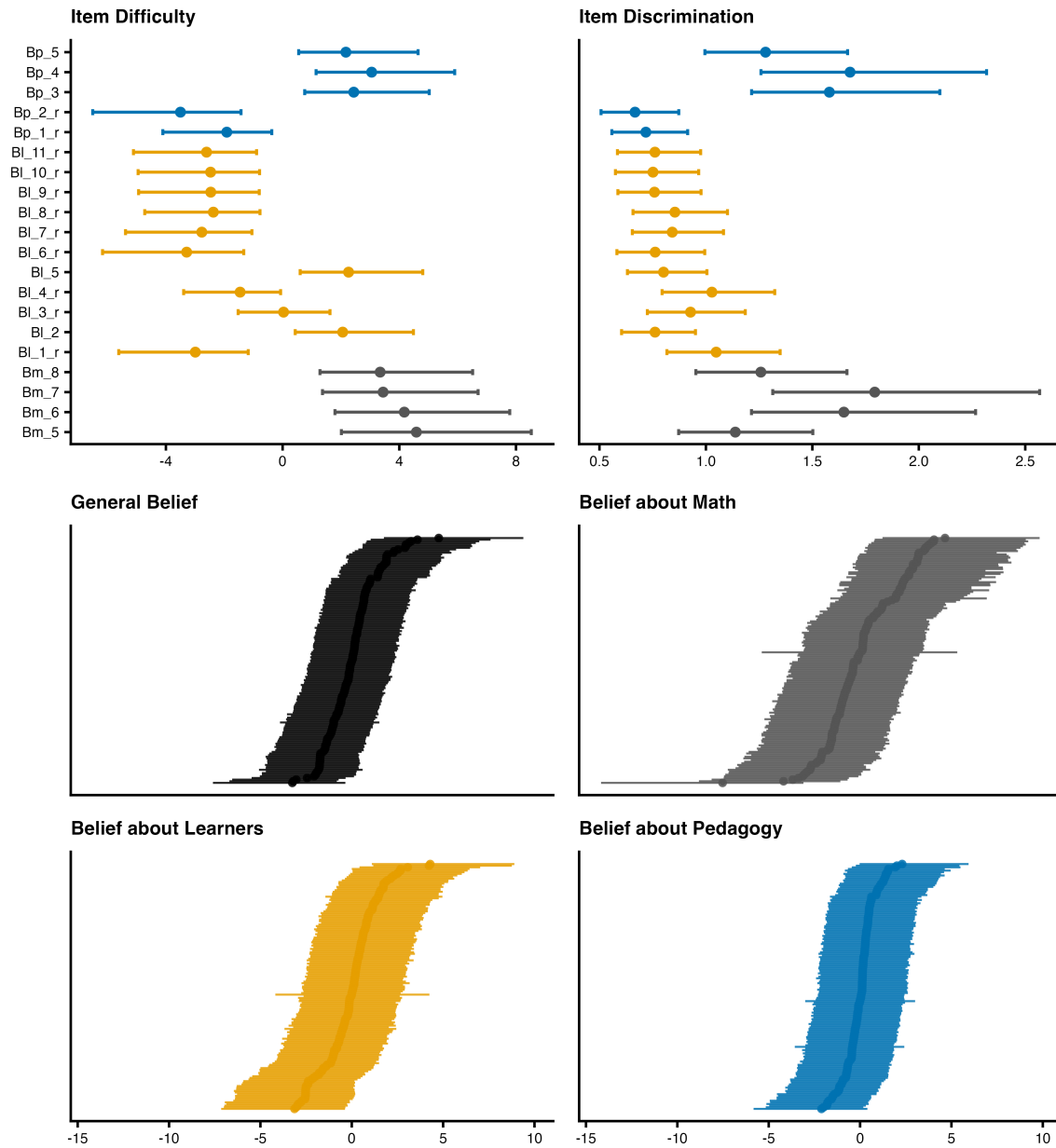
The calibrated comparison shows that the two GRM models are practically indistinguishable in predictive accuracy, but bf_grm_20 receives 100% of the stacking weight. Figure 5 presents the Wright map for bf_grm_20. The top row shows item difficulty and discrimination estimates for the 20 retained items. The bottom four panels display the person parameter estimates for the general belief factor and the three dimension-specific residual factors (mathematics, learners, pedagogy), all plotted on a common scale.

The item difficulty panel shows a well-distributed spread across the latent continuum with no abnormally wide credible intervals, confirming that the removal of the four misfitting items resolved the identification problems observed in Step 3. Discrimination values for the retained items are all above the $\alpha = 0.5$ threshold. The person parameter panels show approximately normal distributions for all four factors, with the general factor exhibiting the widest spread, consistent with its role as the dominant source of shared variance.

Table 5 provides the key parameter estimates. All $\hat{R}$ values are at or below 1.01,

Figure 5

Diagnostic Wright map for the final model (bf_grm_20). Top left: item difficulty with 95% CrIs. Top right: item discrimination (exponentiated scale). Bottom panels: person parameter estimates for the general factor and three dimension-specific factors, sorted by magnitude.



indicating satisfactory convergence.

Table 5

Summary of key parameter estimates from the final model (bf_grm_20). Person parameter SDs are reported for the bi-factor structure; thresholds are population-level intercepts on the cumulative logit scale.

Parameter	Estimate	95% CrI	$\hat{R}$
Person parameters			
SD of general belief	1.76	[0.85, 3.10]	1.00
SD of math-specific belief	2.60	[1.26, 4.47]	1.00
SD of learner-specific belief	1.97	[0.93, 3.39]	1.00
SD of pedagogy-specific belief	1.39	[0.55, 2.59]	1.00
Thresholds			
1: “disagree” to “neutral”	−4.16	[−7.42, −1.89]	1.00
2: “neutral” to “agree”	−0.93	[−2.61, 0.39]	1.01
T3: “agree” to “strongly agree”	5.60	[2.57, 10.06]	1.00

The three thresholds are now in strictly monotonic order with well-separated credible intervals, confirming that the combination of category collapsing (Step 1), multidimensional structure (Step 2), freed discrimination (Step 3), and item removal (Step 4) has fully resolved the threshold anomaly that motivated the iterative refinement process.

Among the person parameters, the math-specific factor has the largest SD (2.60), suggesting that after accounting for a shared equity orientation, teachers vary most in their beliefs about the nature of mathematics. The pedagogy-specific factor has the smallest SD (1.39), indicating relatively less residual variation in beliefs about teaching approaches. The general factor SD (1.76) falls between the two, confirming that all four factors capture meaningful individual differences.

Step 5: Differential item functioning and covariates

Finally, we conduct a DIF analysis on the selected model to ensure that items function consistently across demographic subgroups, a property known as measurement invariance. When an item exhibits differential item functioning (DIF), respondents from different groups who share the same level of the latent trait have systematically different probabilities of endorsing a given category, which can bias group comparisons.

One intuitive approach is to specify $(0 + \text{gender} \mid \text{item})$ alongside gender as a fixed effect. This estimates separate item-level random effects for males and females, giving us distinct posterior distributions of item difficulty for each group:

```
bf_grm_dif <- brm(
  formula = bf(
    response ~ 1 + gender + (0 + gender | item) +
      (1 || id) + (0 + dim_m || id) +
      (0 + dim_l || id) + (0 + dim_p || id),
    disc ~ 1 + (1 | item)
  ),
  data = data_3,
  family = brmsfamily("cumulative", "logit")
)
```

The benefit of this parameterisation is that we can directly inspect how males and females perform on each item. However, determining whether the male-female difference is statistically meaningful is less straightforward, because we do not have a single posterior for the difference itself. Bohn et al. (2025) offers an engineering-level approximation: treating DIF as a diagnostic question and flagging items where the male and female posteriors are non-overlapping. Panel A of Figure 6 shows this diagnostic for all 20 items, and by this criterion all items pass the DIF test. It should be noted, however, that this is an overly optimistic diagnostic: even when two posterior distributions overlap, the difference between them can still be credible.⁴

A more straightforward approach is to model DIF directly by specifying $(1 + \text{gender} \mid \text{item})$, which treats one category as the baseline and estimates the male-minus-female deviation for each item. An item exhibits credible DIF when the 95% CrI for this deviation falls entirely on one side of zero. We extend this to include group variables simultaneously, testing for DIF across both gender and district:

⁴ This is analogous to a situation in frequentist inference where two confidence intervals overlap but the difference between the two means can be nonetheless statistically significant.

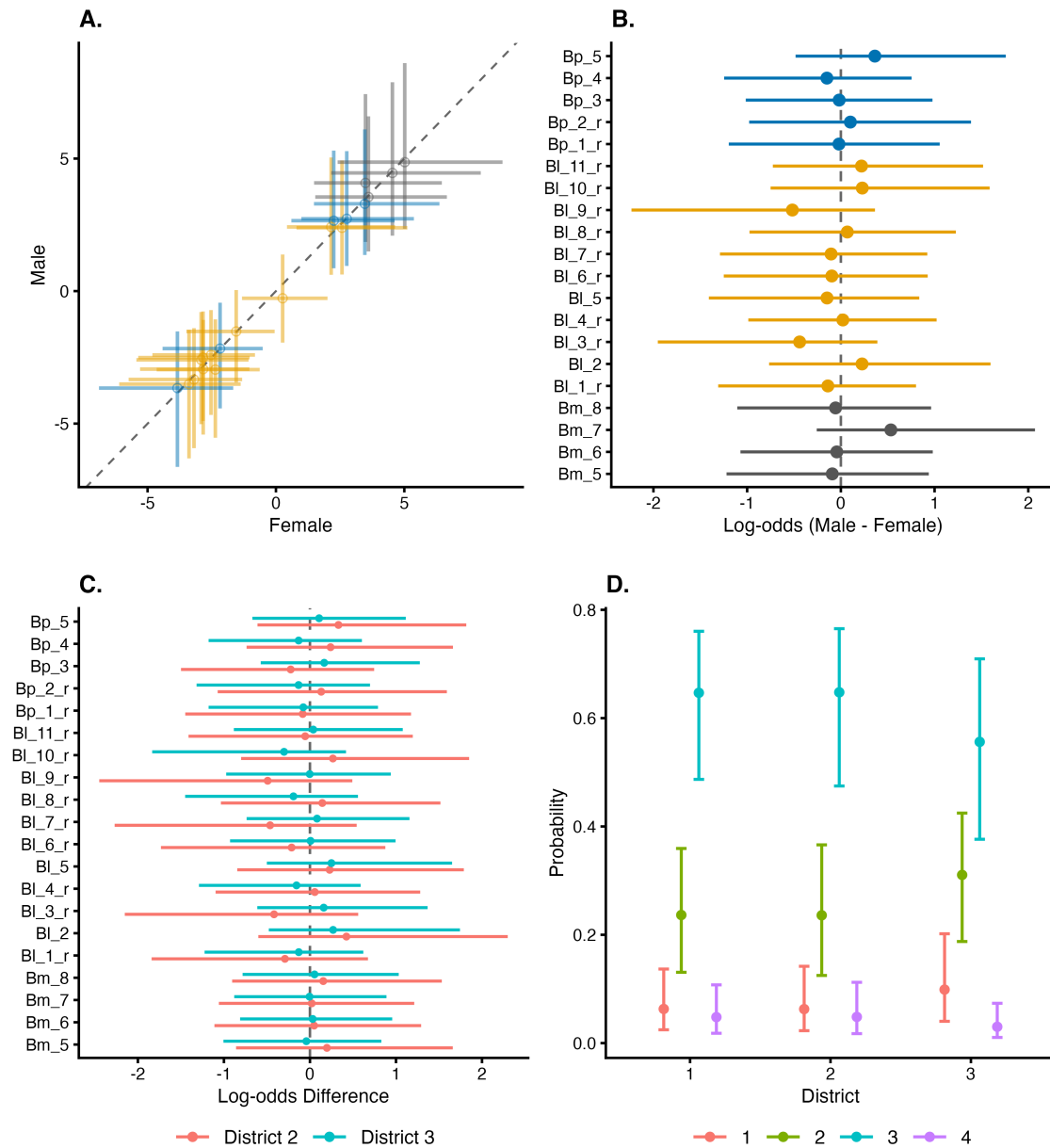
```
bf_grm_dif2 <- brm(
  formula = bf(
    response ~ 1 + gender + District +
      # Forces independent random slopes to aid convergence
      (1 + gender + District || item) +
      (1 || id) + (0 + dim_m || id) +
      (0 + dim_l || id) + (0 + dim_p || id),
    disc ~ 1 + (1 | item)
  ),
  data = data_3,
  family = brmsfamily("cumulative", "logit")
)
```

Figure 6 presents the results. Panel B shows the item-level gender DIF effects (log-odds difference, male minus female) from `bf_grm_dif2`, and all 20 items have 95% CrIs that include zero; Panel C shows the similar results for district (Districts 2 and 3 relative to District 1 as baseline), and again no item exhibits a credible DIF effect.

Because covariates are included directly in the measurement model, we simultaneously obtain group-level effects alongside the item-level DIF tests. Several notable population-level (fixed) effects emerge from `bf_grm_dif2`. The gender effect is strictly credible, showing that male teachers hold less inclusive beliefs (-1.20 , 95% CrI $[-2.62, -0.04]$). The district effect, while its 95% CrI slightly overlaps zero for District 3 (-1.29 , 95% CrI $[-2.89, 0.03]$), remains highly noteworthy. The vast majority of the posterior distribution lies below zero, indicating a strong downward trend in inclusive beliefs for teachers in lower-SES districts. As visualized in Panel D, this manifests in a systematic shift toward lower response categories.

Figure 6

Differential item functioning and covariate analysis. Panel A: posterior overlap diagnostic for gender bf_grm_dif . Panel B: item-level gender DIF effects from bf_grm_dif2 . Panel C: item-level district DIF effects. Panel D: Estimated category response probabilities by district.



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Appendix A

Model specifications

This appendix provides detailed mathematical specifications of the ordinal IRT models used in this study. All models are extensions of IRT for ordered categorical responses, differing in their link functions, threshold parameterisations, and discrimination constraints. Implementation details in brms are provided in the main text and companion repository.

Ordinal Link Functions and Threshold Parameterisations

Rating Scale Model (RSM)

The RSM uses the adjacent-category logit link with shared thresholds. The probability of person p responding in category c ($c = 1, \dots, C$) to item i is:

$$P(Y_{pi} = c) = \frac{\exp \sum_{j=1}^c (\theta_p + \xi_i - \tau_j)}{\sum_{r=1}^C \exp \sum_{j=1}^r (\theta_p + \xi_i - \tau_j)}$$

where θ_p is the latent trait of person p , ξ_i is the item-specific location (random intercept; more negative = harder to endorse), and τ_j is the j -th threshold parameter. A key constraint of the RSM is that thresholds τ_j are shared across all items, assuming equal category spacing across the instrument.

Generalised Rating Scale Model (GRSM)

Introducing item-specific discrimination parameters α_i to the RSM yields the GRSM:

$$P(Y_{pi} = c) = \frac{\exp \sum_{j=1}^c \alpha_i (\theta_p + \xi_i - \tau_j)}{\sum_{r=1}^C \exp \sum_{j=1}^r \alpha_i (\theta_p + \xi_i - \tau_j)}$$

Higher α_i values indicate items that more sharply differentiate between respondents at different trait levels. Thresholds remain shared across items.

Generalised Partial Credit Model (GPCM)

The GPCM combines item-specific discrimination with item-specific thresholds $\tau_{i,j}$:

$$P(Y_{pi} = c) = \frac{\exp \sum_{j=1}^c \alpha_i(\theta_p + \xi_i - \tau_{i,j})}{\sum_{r=1}^C \exp \sum_{j=1}^r \alpha_i(\theta_p + \xi_i - \tau_{i,j})}$$

Unlike the GRSM, which constrains all items to share the same threshold spacing, the GPCM allows each item to have its own set of threshold parameters. This provides the most flexible parameterisation within the adjacent-category family, but requires estimating substantially more parameters, which can compromise stability with small samples.

Graded Response Model (GRM)

The GRM uses the cumulative logit link rather than the adjacent-category logit. In its general form (Samejima, 2016), the GRM allows fully item-specific thresholds. The implementation used in this study is a constrained variant in which threshold spacing is shared across items, with item differences captured through a location parameter ξ_i that shifts all thresholds equally. The probability of responding in category c is defined as:

$$P(Y_{pi} = c) = F(\alpha_i(\eta_{pi} - \tau_{c-1})) - F(\alpha_i(\eta_{pi} - \tau_c))$$

where $\eta_{pi} = \theta_p + \xi_i$ is the person-by-item location, $F(x) = \frac{1}{1+\exp(-x)}$ is the logistic CDF, α_i is item discrimination, and the thresholds are ordered: $\tau_1 < \dots < \tau_{C-1}$, with boundary conditions $\tau_0 = -\infty$ and $\tau_C = +\infty$. The thresholds τ_c are shared population-level parameters; item i 's effective thresholds are $\tau_c - \xi_i$, preserving equal inter-threshold spacing across items.

The GRM directly models the cumulative probability $P(Y_{pi} \geq c)$, in contrast to the adjacent-category models which model the odds of each category relative to its neighbour.

Summary of Model Families

Dimensional Structures

All models in this study are implemented as generalised linear mixed models (GLMMs) in brms, following the framework described in De Boeck et al. (2011) and Bürkner (2021).

Table A1

Summary of ordinal IRT model families used in this study. RSM, GRSM, and GRM share threshold spacing across items; item differences enter only through the location parameter ξ_i . GPCM allows fully item-specific thresholds. The generalised forms (GRSM, GPCM, GRM) estimate item-specific discrimination.

Model	Link	Thresholds	Discrimination
RSM	Adjacent-category	Shared	Equal ($\alpha = 1$)
GRSM	Adjacent-category	Shared	Item-specific
GPCM	Adjacent-category	Item-specific	Item-specific
GRM	Cumulative	Shared (equal spacing)	Item-specific

Unidimensional IRT

$$\eta_{pi} = \beta_0 + \xi_i + \theta_p$$

where β_0 is the fixed intercept, $\xi_i \sim \mathcal{N}(0, \sigma_{\text{item}}^2)$ is the item-specific random intercept, and $\theta_p \sim \mathcal{N}(0, \sigma_{\theta}^2)$ is the person-specific random effect.

Correlated-Traits MIRT (CT-MIRT)

Each person p has a vector of dimension-specific traits:

$$\boldsymbol{\theta}_p = (\theta_{pM}, \theta_{pL}, \theta_{pP})^\top \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

where $\boldsymbol{\Sigma}$ is a 3×3 covariance matrix. The linear predictor is:

$$\eta_{pi} = \beta_0 + \xi_i + \sum_{d \in \{M, L, P\}} \mathbb{I}_{id} \theta_{pd}$$

where $\mathbb{I}_{id} = 1$ if item i belongs to dimension d and 0 otherwise (each item loads on exactly one dimension). The unrestricted covariance matrix $\boldsymbol{\Sigma}$ allows the three latent dimensions to correlate freely, reflecting the possibility that the dimensions are distinct but related constructs.

Bi-Factor MIRT (BF-MIRT)

A general factor θ_p is added alongside dimension-specific factors θ_{pd}^s :

$$\eta_{pi} = \beta_0 + \xi_i + \theta_p + \sum_{d \in \{M, L, P\}} \mathbb{I}_{id} \theta_{pd}^s$$

where $\theta_p \sim \mathcal{N}(0, \sigma_\theta^2)$ and $\theta_{pd}^s \sim \mathcal{N}(0, \sigma_d^2)$. The general and specific factors are constrained to be orthogonal:

$$\text{Cov}(\theta_p, \theta_{pd}^s) = 0 \quad \forall d$$

This constraint ensures that the specific factors capture only residual variation within each dimension after accounting for the general factor. The distinction from CT-MIRT is structural: the correlated-traits model permits free correlations among dimension-specific traits but includes no general factor, whereas the bi-factor model decomposes variance into a general component shared across all items and orthogonal dimension-specific residuals.

Prior Distributions

Weakly informative priors are used throughout, following recommendations in Bürkner (2021):

Table A2

Prior distributions for all model parameters.

Parameter	Prior	Rationale
Person and item SDs ($\sigma_\theta, \sigma_{\text{item}}$)	$\mathcal{N}(0, 3)$	Half-normal (truncated at 0); allows substantial variation while regularising.
Log-discrimination SD	$\mathcal{N}(0, 1)$	Covers discrimination values roughly in $[0.14, 7.4]$.
Correlation matrix (CT-MIRT)	LKJ(2)	Mildly favours smaller correlations; $\eta > 1$ concentrates density near the identity matrix.

Appendix B

Instrument items

Teachers were asked to report their endorsement of each statement on a 5-point Likert scale from “strongly disagree” (1) to “strongly agree” (5). Items ending with _r are reverse-coded: responses are recoded prior to analysis so that higher scores consistently reflect stronger endorsement of inclusive and equity-oriented beliefs.

Bm: Beliefs About Mathematics (8 Items)

Table B1

Beliefs about mathematics items.

Item	Statement
Bm_1_r	Mathematics is a collection of rules and procedures.
Bm_2_r	Mathematics involves remembering and applying definitions, formulas, mathematical facts and procedures.
Bm_3_r	To do mathematics requires much practice of correct application of routines.
Bm_4_r	When solving mathematical tasks you need to know the correct procedure in advance.
Bm_5	Mathematics involves creativity and new ideas.
Bm_6	Many aspects of mathematics have practical relevance.
Bm_7	Mathematics helps solve everyday problems and tasks.
Bm_8	In mathematics many things can be discovered and tried out by oneself.

Bl: Beliefs About Learners (11 Items)**Table B2***Beliefs about learners items.*

Item	Statement
Bl_1_r	Students must be able to solve mathematics problems quickly.
Bl_2	Students must be able to understand why the answer is correct.
Bl_3_r	Students learn mathematics best by attending to the teacher's standard explanations.
Bl_4_r	For students, non-standard procedures can interfere with learning the correct procedure.
Bl_5	Students can figure out a way to solve mathematical problems without a teacher's help.
Bl_6_r	Only the more able students can participate in mathematics activities.
Bl_7_r	To be good at mathematics one needs to be talented.
Bl_8_r	Among other forms of talent, mathematical talent is relatively fixed.
Bl_9_r	Boys tend to be more talented in mathematics than girls.
Bl_10_r	Attention to mathematically weak students can interfere with the learning experience of other students.
Bl_11_r	Attention to mathematically talented students can interfere with the learning experience of other students.

Bp: Beliefs About Pedagogy (5 Items)**Table B3***Beliefs about pedagogy items.*

Item	Statement
Bp_1_r	Innovative pedagogy (e.g., project-based learning, flipped classroom) is not worth the time and expense.
Bp_2_r	Innovative pedagogy (e.g., project-based learning, flipped classroom) should only play a minor, supplementary role in the mathematics classroom.
Bp_3	I'm continuously seeking better ways to teach mathematics.
Bp_4	Teachers should help students using mathematics to critically analyse the world.
Bp_5	Discussion of social issues should be involved in the maths classroom.